READY FOR FINAL VERIFICATION

Campus Finance

Web Programming | Milestone 1 | Frontend Application

Team name: Campus Finance **Project idea:** Campus Finance & Cafeteria Management

1. Team members

#	Member	University ID	University email
1	Eyad	16009022	eyad.diab@student.giu-uni.de
2	Yassin	16004778	Yasin.ebaid@student.giu-uni.de
3	Ali	16001570	aley.azmi@student.giu-uni.de

2. Application description

Campus Finance is a responsive Angular frontend prototype combining university tuition invoicing with student cafeteria payments and departmental finance workflows. Each user signs in to view features appropriate for their role. All data and financial actions are simulated for Milestone 1; there is no backend, database, payment gateway or real banking integration.

3. Implemented features by role

Student
Tuition invoices and history; digital wallet; food catalog, cart, checkout; meal plans.

Accounting / Admin
Create and record demo invoice payments; payroll; expense and income reporting.

Cafeteria staff
See in-session student orders and update their status.

Shared access
Home, signup, login, profile, role-aware links and guards.

4. Frontend design and interaction

Standalone Angular components are loaded lazily through the router; Angular reactive forms validate signup and invoice inputs. Angular signals and injected services coordinate app state. Tailwind CSS and SCSS define responsive layouts. Chart.js visualizes department-budget totals and categories.

Data storage & user experience

Selecting storage based on privacy, persistence and convenience

5. Storage strategy and justification

Mechanism	Stored data	Why / UX effect
sessionStorage	Demo signed-in user; temporary signup username, optional small profile image, and fictional card last-four digits.	Restores a user's demo profile across a refresh in the same tab, while clearing on tab close. The password is not saved by registration.
localStorage	Payroll employee/pay records and budget income/expense entries.	Retains non-production finance examples after refresh, making repeated demos quicker. Accessible only to this browser profile/device.
Cookie	Non-sensitive food-preference setting.	Remembers the preference across visits for 30 days. SameSite=Lax; no password or payment credential is stored in cookies.
In-memory	Wallet balance changes, new invoices, meal subscriptions and cafeteria orders.	Fast reactive UI with no server dependency; changes reset on full reload. Not durable or shared across devices.

6. Privacy, security and UX considerations

- Client-side guards restrict visible routes by role (student, accounting, admin, cafeteria staff). This is a UI demonstration, not production-grade authorization.
- Registration collects a *fictional* card's last four digits only. Never enter real card details, expiry or CVV. No real payments are processed.
- The optional profile image is restricted to a small PNG, JPEG or WebP file and stored only in the session demo.
- Cookie preferences are deliberately non-sensitive; role/user state is kept separately in the tab session. Browser storage can be inspected by the device's user.
- Payroll and budget local data saves time after refresh, but the lack of server synchronization limits collaboration and auditing.

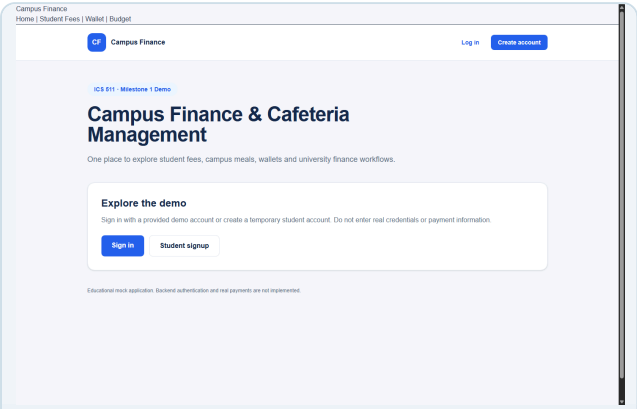
7. Testing and limitations

On 12 November 2026, the compliance branch passed the Angular production build and **142/142 Jasmine/Karma tests** locally. The tested code was subsequently merged into `master` (merge commit e7cd1ac). These are local test results rather than CI or production certification. Chrome browser demonstrations should verify responsive views, signup/profile persistence and role access.

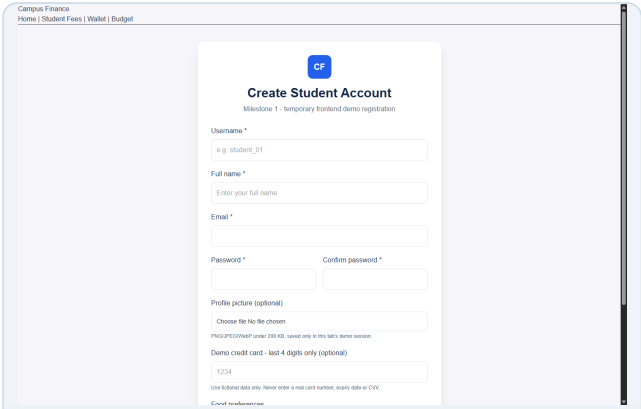
Known limitations: mocked authentication; simulated wallet/tuition payments; in-memory order and invoice history; no automatic cancellation refund; backend and database intentionally deferred to future milestones.

Screenshots from the application

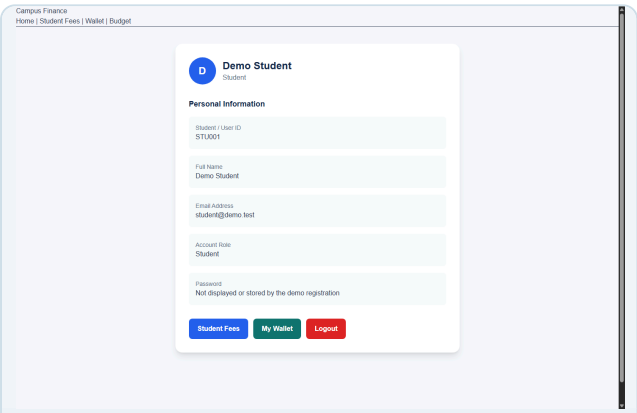
Actual Chrome screenshots captured locally - not mockups



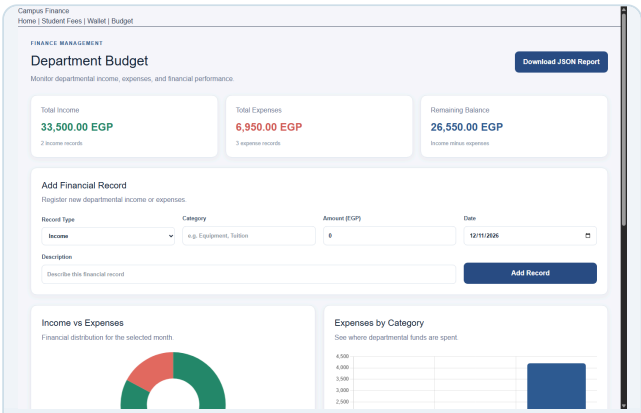
Home / entry page



Student sign-up form



Profile with sign-up data



Department budget and charts

8. Sample walkthrough

Student: Open Home, select Signup, enter a username and other requested demo fields, inspect the Profile page and then use Student Fees, Wallet, Food Ordering and Meal Plans. **Admin/Accounting:** Sign in with seeded demo credentials and view Payroll, Department Budget and their charts. **Staff:** Review cafeteria orders created during the same running app session.

Source code: https://github.com/eyaddiab6/campus-finance | Milestone deadline: 14 Nov 2026 | Submission link: [ADD AFTER UPLOADING]